Happiness data

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Packages used

```
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.1.4     v readr     2.1.5
## v forcats   1.0.0     v stringr   1.5.1
## v ggplot2   3.5.1     v tibble    3.2.1
## v lubridate 1.9.3     v tidyr     1.3.1
## v purrr     1.0.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```
library(maps)
```

```
## Warning: package 'maps' was built under R version 4.4.2
```

```
##
## Attaching package: 'maps'
##
## The following object is masked from 'package:purrr':
##
##     map
```

Import data set

```
happiness <- read.csv("world-happiness-report-2021.csv")
```

Subset initial date, final date, area in km squared, and country name

```
happiness <- subset(happiness, select = c(Country.name, Regional.indicator, Ladder.score, Logged.GDP.pe
head(happiness)
```

```
##   Country.name Regional.indicator Ladder.score Logged.GDP.per.capita
## 1      Finland     Western Europe        7.842                10.775
## 2      Denmark     Western Europe        7.620                10.933
## 3  Switzerland     Western Europe        7.571                11.117
```

```
## 4      Iceland     Western Europe        7.554                10.878
## 5  Netherlands     Western Europe        7.464                10.932
## 6       Norway     Western Europe        7.392                11.053
##   Healthy.life.expectancy
## 1                    72.0
## 2                    72.7
## 3                    74.4
## 4                    73.0
## 5                    72.4
## 6                    73.3
```

Check data types, make region a factor, and rename America

```
happiness$Regional.indicator <- as.factor(happiness$Regional.indicator)
happiness[19,1] <- "USA"
str(happiness)
```

```
## 'data.frame':    149 obs. of  5 variables:
##  $ Country.name           : chr  "Finland" "Denmark" "Switzerland" "Iceland" ...
##  $ Regional.indicator     : Factor w/ 10 levels "Central and Eastern Europe",..: 10 10 10 10 10 10 10
##  $ Ladder.score           : num  7.84 7.62 7.57 7.55 7.46 ...
##  $ Logged.GDP.per.capita  : num  10.8 10.9 11.1 10.9 10.9 ...
##  $ Healthy.life.expectancy: num  72 72.7 74.4 73 72.4 73.3 72.7 72.6 73.4 73.3 ...
```

Set up map

```
world.map <- map_data("world")
happiness.map <- merge (world.map, happiness, by.x = "region", by.y = "Country.name")
happiness.map <- arrange (happiness.map, group, order )
```

Map of happiness scores

```
ggplot (happiness.map , aes(long, lat, group = group, fill = Ladder.score)) +
  geom_polygon(color = " black ") +
  scale_fill_gradient(low = "white", high = "darkgreen" )
```

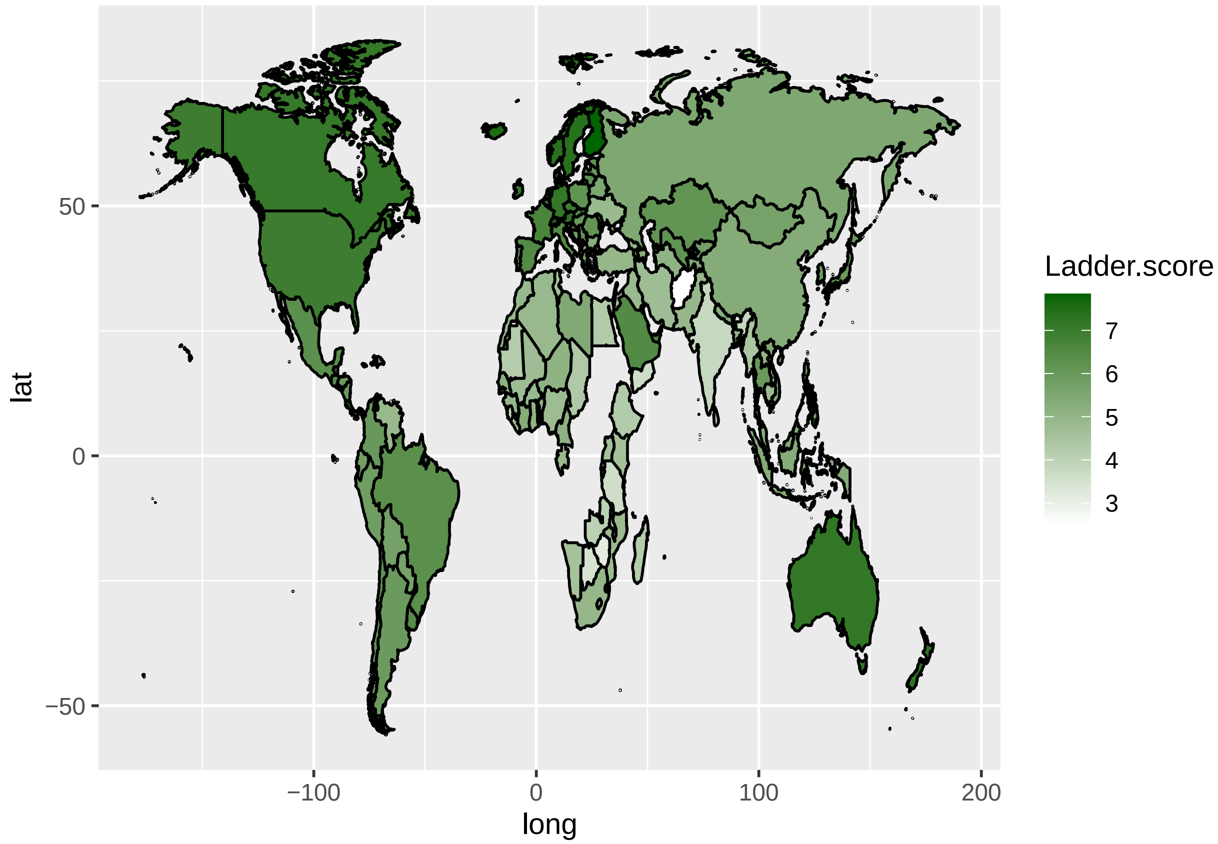


Chart of GDP per Capita vs Happiness score

```r
ggplot (happiness, aes(x = Logged.GDP.per.capita, y = Ladder.score)) +
  geom_point() +
  geom_smooth(method = "lm", se=F) +
  xlab("GDP per Capita") +
  ylab("Happiness Score")
```

```
## `geom_smooth()` using formula = 'y ~ x'
```

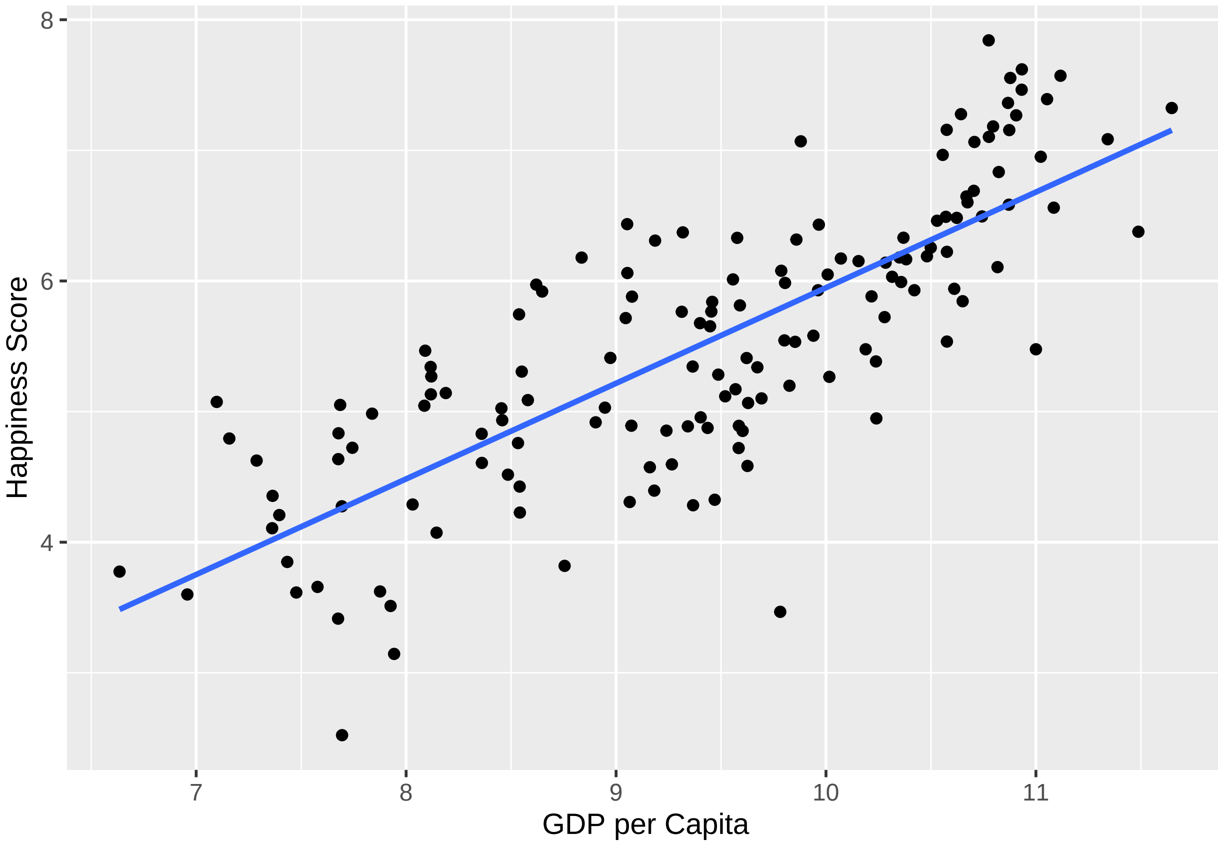
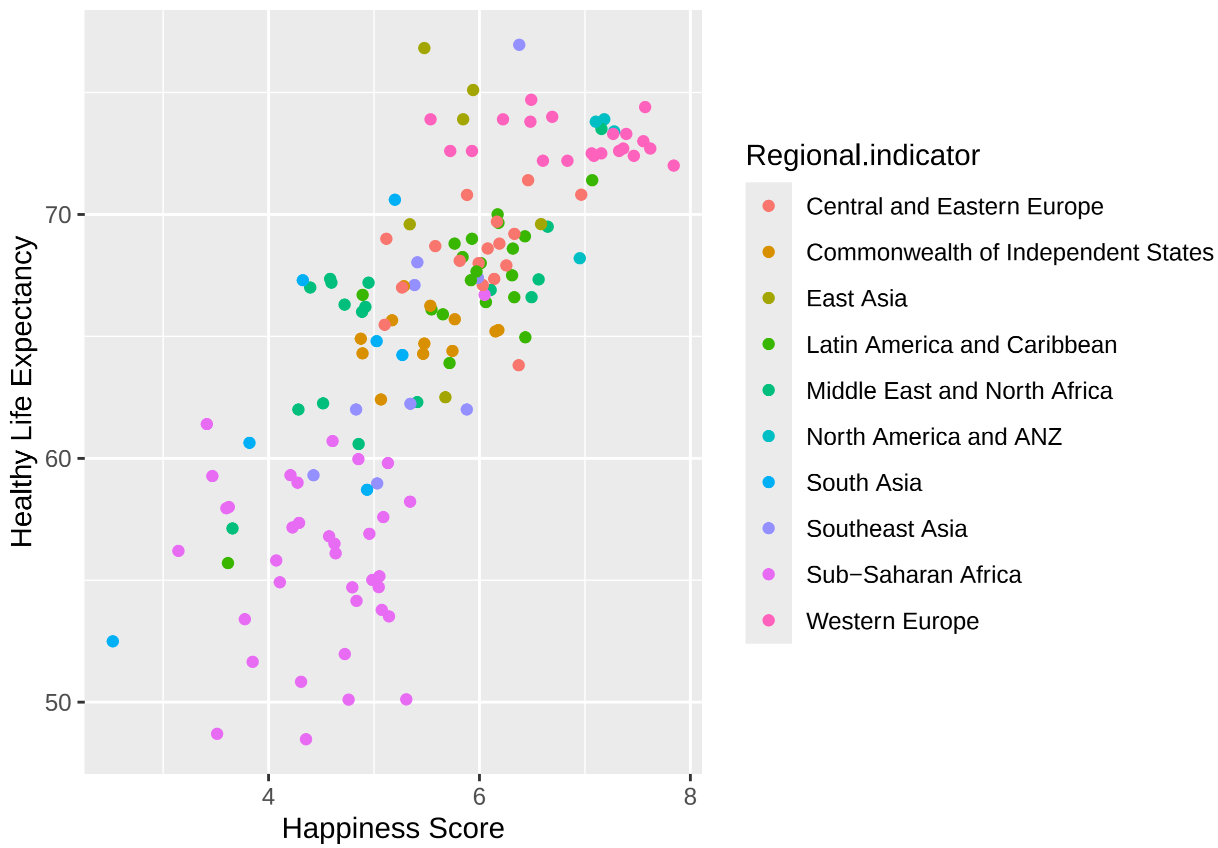


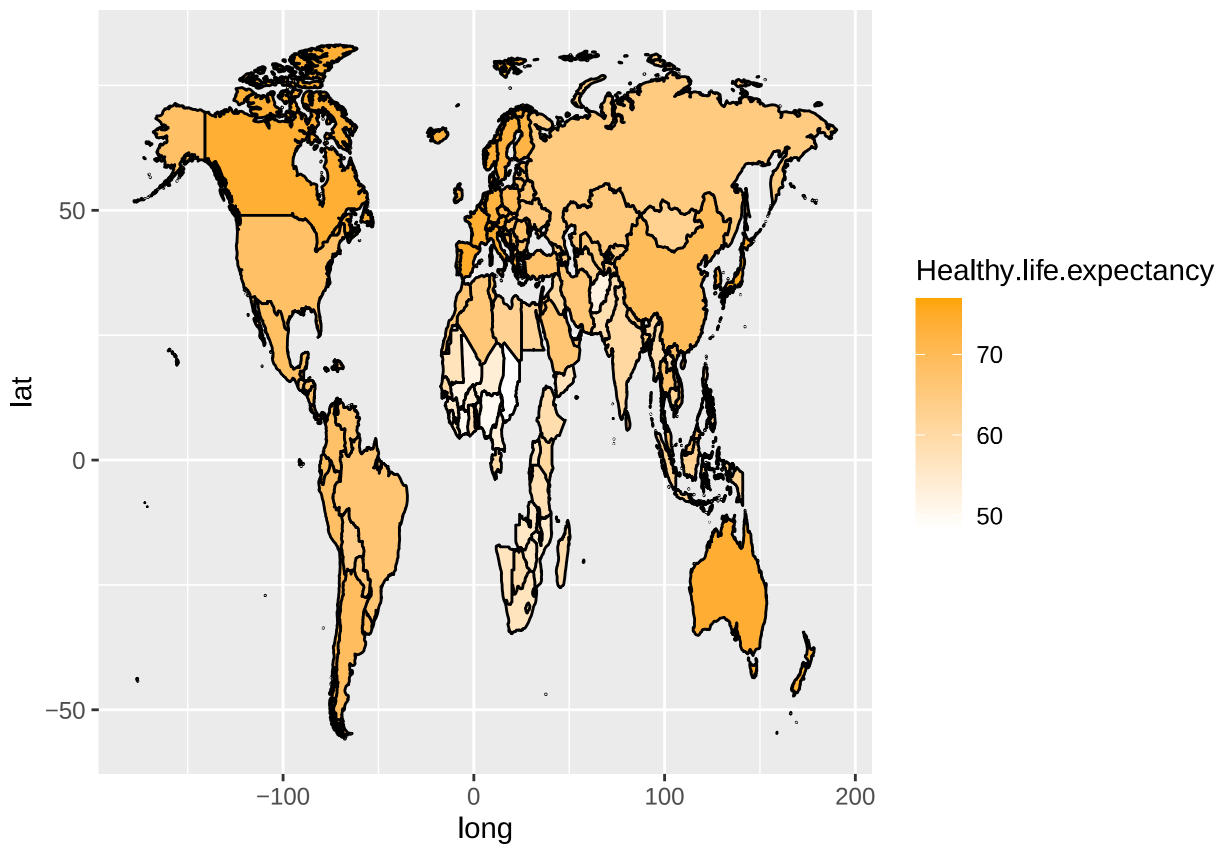
Chart of happiness level vs healthy life expectancy

```
ggplot (happiness, aes(x = Ladder.score, y = Healthy.life.expectancy)) +
  geom_point(aes(color = Regional.indicator)) +
  ylab("Healthy Life Expectancy") +
  xlab("Happiness Score")
```



Map of healthy life expectancy

```
ggplot (happiness.map , aes(long, lat, group = group, fill = Healthy.life.expectancy)) +
  geom_polygon(color = " black ") +
  scale_fill_gradient(low = "white", high = "Orange" )
```



Healthy.life.expectancy
70
60
50
lat
50
0
−50
long
−100
0
100
200